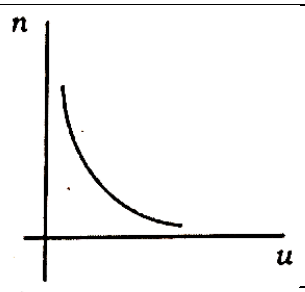
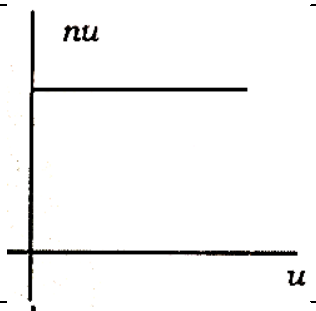
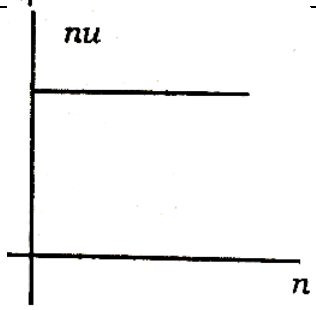
Question	Light year is a unit of
Type	multiple choice
Option	Time
Option	Mass
Option	Distance
Option	Energy
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following is the smallest unit
Type	multiple choice
Option	Millimetre
Option	Angstrom
Option	Fermi
Option	Metre
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The SI unit of momentum is
Type	multiple choice
Option	$\frac{kg}{m}$
Option	$\frac{kg.m}{\sec}$
Option	$\frac{kg.m^2}{\sec}$
Option	$kg \times Newton$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Candela is the unit of
Type	multiple choice
Option	Electric intensity
Option	Luminous Intensity
Option	Sound Intensity
Option	None of these
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The value of Gravitational constant G in MKS system is $6.67 \times 10^{-11} N - m^2 / kg^2$. What will be its value in CGS system?
Type	multiple choice
Option	$6.67 \times 10^{-10} dyne - cm^2 / g^2$
Option	$6.67 \times 10^{-8}.dyne - cm^2 / g^2$
Option	$6.67 \times 10^{-9} dyne - cm^2 / g^2$
Option	$6.67 \times 10^{-7} dyne - cm^2 / g^2$

Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following does not have the same unit as others?
Type	multiple choice
Option	Watt-s
Option	kilowatt-hour
Option	eV
Option	J-s
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The density of mercury is 13600 kg m^{-3}. Its value in CGS system will be:
Type	multiple choice
Option	$113.6\,g\,cm^{-3}$
Option	$1360\,g\,cm^{-3}$
Option	$136\,g\,cm^{-3}$
Option	$1.36\,g\,cm^{-3}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The pressure of 10^6 dyne/cm² is equivalent to
Type	multiple choice
Option	$10^5\,N/m^2$
Option	$10^6\,N/m^2$
Option	$10^7\,N/m^2$
Option	$10^8\,N/m^2$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If u_1 and u_2 are the units selected in two systems of measurement and n_1 and n_2 are their numerical values, then
Type	multiple choice
Option	$n_1u_1 = n_2u_2$
Option	$n_1u_1 + n_2u_2 = 0$
Option	$n_1u_2 = n_1u_2$
Option	$(n_1 + u_1) = (n_2 + u_2)$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following graph is correct: n = magnitude of measurement & u = unit of measurement.
Type	multiple choice

Option	
Option	
Option	
Option	All of these
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Planck's constant has the dimensions (unit) of
Type	multiple choice
Option	Energy
Option	Linear momentum
Option	Work
Option	Angular momentum
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	The dimensions of Energy are
Type	multiple choice
Option	ML^2T^{-2}
Option	MLT^{-2}
Option	ML^2T^{-1}
Option	ML^2T^{-3}
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which pair has the same dimensions?
Type	multiple choice
Option	Work and power
Option	Density and relative density

Option	Momentum and impulse
Option	Stress and strain
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	If L and R are respectively the inductance and resistance, then the dimensions of $\frac{L}{R}$ will be
Type	multiple choice
Option	$M^0L^0T^{-1}$
Option	M^0LT^0
Option	M^0L^0T
Option	Can not be represented in terms of M, L and T
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The dimensional formula of angular velocity is
Type	multiple choice
Option	$M^0L^0T^{-1}$
Option	MLT^{-1}
Option	$M^0L^0T^1$
Option	ML^0T^{-2}
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If L denotes the inductance of an inductor through which a current, I, is flowing, the dimensions of Li^2 are
Type	multiple choice
Option	ML^2T^2
Option	Not expressible in MLT
Option	ML^2T^{-2}
Option	$M^2L^2T^{-2}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	Dimensional formula of velocity of sound is
Type	multiple choice
Option	M^0LT^{-2}
Option	LT^0
Option	M^0LT^{-1}
Option	$M^0L^{-1}T^{-1}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The dimensions of coefficient of thermal conductivity is
Type	multiple choice

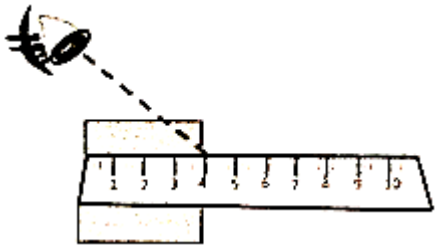
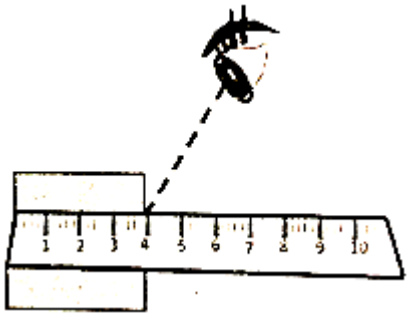
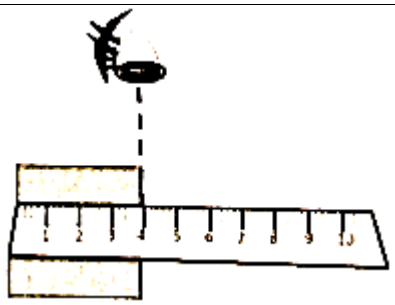
Option	$ML^2T^{-2}K^{-1}$
Option	$MLT^{-3}K^{-1}$
Option	$MLT^{-2}K^{-1}$
Option	$MLT^{-3}K$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	μ_0 and ε_0 denote the permeability and permittivity of free space, the dimensions of $\mu_0\varepsilon_0$ are
Type	multiple choice
Option	LT^{-1}
Option	$L^{-2}T^2$
Option	$M^{-1}L^{-3}Q^2T^2$
Option	$M^{-1}L^{-3}I^2T^2$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The dimensions of physical quantity X in the equation Force $= \dfrac{x}{Density}$ is given by
Type	multiple choice
Option	$M^1L^4T^{-2}$
Option	$M^2M^{-2}T^{-1}$
Option	$M^2L^{-2}T^{-2}$
Option	$M^1L^{-2}T^{-1}$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A force F is given by $F = at + bt^2$, where t is time. What are the dimensions of a and b
Type	multiple choice
Option	$[MLT^{-3}]$ and $[ML^2T^{-4}]$
Option	$[MLT^{-3}]$ and $[MLT^{-4}]$
Option	$[MLT^{-1}]$ and $[MLT^0]$
Option	$[MLT^{-4}]$ and $[MLT^1]$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If the power of a body is given by $P = xv$, where v is the velocity of the body, then the dimension of x is
Type	multiple choice
Option	$[\mathrm{ML}T^{-2}]$
Option	$[\mathrm{ML}^2\mathrm{T}^{-2}]$
Option	$[\mathrm{ML}T^2]$

Option	$[M^2 L T^2]$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	If y represent pressure and x represents velocity gradient, then the dimensions of $\frac{d^2y}{dx^2}$ are
Type	multiple choice
Option	$[ML^{-1}T^{-2}]$
Option	$[M^2L^{-2}T^{-2}]$
Option	$[ML^{-1}T^{0}]$
Option	$[M^2L^{-2}T^{-4}]$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A physical quantity P is given by the relation. $P = P_0 e^{-\alpha t^2}$. If t denotes the time, the dimensions of constant α is
Type	multiple choice
Option	$[T]$
Option	$[T^2]$
Option	$[T^{-1}]$
Option	$[\dot{T}^{-2}]$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	If $\alpha = \frac{F}{v^2}\sin\beta t$, find dimensions of β and α. Here $v =$ velocity, $F =$ force and $t =$ time.
Type	multiple choice
Option	$[T^{-1}]$ and $[ML^{-1}]$
Option	$[T^{-2}]$ and $[ML^{-1}]$
Option	$[T^{-1}]$ and $[M^{-1}L]$
Option	$[T]$ and $[ML^{-1}]$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	An equation is given here $\left(P + \frac{a}{v^2}\right) = b\frac{\theta}{v}$ where $P =$ Pressure, $V =$ Volume and $\theta =$ Absolute temperature. If a and b are constants, the dimensions of a will be
Type	multiple choice
Option	$[ML^{-5}T^{-1}]$

Option	$[ML^5T^1]$
Option	$[ML^5T^{-2}]$
Option	$[M^{-1}L^5T^2]$
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity v of a particle at time t is given by $v = at + \frac{b}{t+c}$, where a, b and c are constants. The dimensions of a, b and c are
Type	multiple choice
Option	$[L]$, $[LT]$ and $[LT^{-2}]$
Option	$[LT^{-2}]$, $[L]$ and $[T]$
Option	$[L^2]$, $[T]$ and $[LT^{-2}]$
Option	$[LT^{-2}]$, $[LT]$ and $[L]$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	The velocity of a free fall body is expressed as $g^p h^q$ where g is acceleration due to gravity and h is the height. The value of p and q are
Type	multiple choice
Option	$1, \frac{1}{2}$
Option	$\frac{1}{2}, \frac{1}{2}$
Option	$\frac{1}{2}, 1$
Option	1,1
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	If $A = \frac{\rho v d}{\mu}$, where ρ is the density, v is the velocity, d is the diameter and μ is the dynamic viscosity, then find the dimension of A. (Hint: Dimension of dynamic viscosity $[\mu] = [ML^{-1}T^{-2}]$),
Type	multiple choice
Option	$[ML^{-1}T]$
Option	$[ML^{-2}T^{-1}]$
Option	$[ML^{-3}T^{-1}]$
Option	Dimensionless
Answer	4
Solution	
Positive Marks	4
Negative Marks	1

Question	
Type	If P represents radiation pressure, c represents speed of light and Q represents radiation energy striking a unit area per second, then non-zero integers x, y and z such that $P^x Q^y c^z$ is dimensionless are
Option	multiple choice
Option	$x=1, y=1, z=-1$
Option	$x=1, y=-1, z=1$
Option	$x=-1, y=1, z=1$
Answer	$x=1, y=1, z=1$
Solution	2
Positive Marks	4
Negative Marks	1
Question	The quantity A and B are related by the relation, $m = A/B$, where m is the linear mass density and A is the force. The dimensions of B are of
Type	multiple choice
Option	Pressure
Option	Work
Option	Latent heat
Option	None of the above
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The physical quantity which has dimensional formula as that of $\frac{Energy}{Mass \times Length}$ is
Type	multiple choice
Option	Force
Option	Power
Option	Pressure
Option	Acceleration
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	In a system of units if force (F), acceleration (A) and time (T) are taken as fundamental units then the dimensional formula of energy is
Type	multiple choice
Option	FA^2T
Option	FAT^2
Option	F^2AT
Option	FAT
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	Given that v is speed, r is the radius and g is the acceleration due to gravity. Which of the following is dimensionless
Type	multiple choice
Option	v^2/rg

Option	$v^2 r / g$
Option	$v^2 g / r$
Option	$v^2 r$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Using the principle of homogeneity of dimensions, which of the following is correct?
Type	multiple choice
Option	$T^2 = \frac{4\pi^2 r^3}{GM}$
Option	$T^2 = 4\pi^2 r^2$
Option	$T^2 = \frac{4\pi^2 r^3}{G}$
Option	$T = \frac{4\pi^2 r^3}{G}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	In a measurement, both positive and negative errors are found to occur with equal probability. This type of error is known as
Type	multiple choice
Option	Proportional error
Option	Systematic error
Option	Determinate error
Option	Random error
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	A student measures the time period of 100 oscillations of a simple pendulum four times. That data set is 90 s, 91 s, 95 s and 92 s. If the minimum division in the measuring clock is 1 s, then the reported mean time should be
Type	multiple choice
Option	$92 \pm 5.0\text{s}$
Option	$92 \pm 1.8\text{s}$
Option	$92 \pm 3\text{s}$
Option	$92 \pm 2\text{s}$
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	What is the smallest change in the input signal that can be detected by an instrument called?
Type	multiple choice
Option	Accuracy
Option	Precision
Option	Resolution (Least count)

Option	Sensitivity
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	The position of the eye is very important for taking measurements. There might be an error in the reading, if our eye position is not proper. This error is known as error
Type	multiple choice
Option	Parallax
Option	Zero
Option	Instrumental
Option	Both (a) and (b) above.
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	Which of the following should be the proper position of the eye for taking the reading?
Type	multiple choice
Option	
Option	
Option	
Option	
Answer	3
Solution	
Positive Marks	4
Negative Marks	1
Question	A broken scale is used to measure the length of a stick, and the reading is

	directly noted down by looking at the scale from a position directly above it. Which of the following is correct?
Type	multiple choice
Option	The measured length is correct.
Option	The measured length is correct because of parallax error
Option	The measured length is incorrect because of parallax error.
Option	The measured length is incorrect because of zero error.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1
Question	Planck's constant (h), speed of light in vacuum (c) and Newton's gravitational constant (G) are three fundamental constants. Which of the following combinations of these has the dimension of length?
Type	multiple choice
Option	$\frac{\sqrt{hG}}{c^2}$
Option	$\frac{\sqrt{hG}}{c^{\frac{5}{2}}}$
Option	$\sqrt{\frac{hc}{G}}$
Option	$\sqrt{Gc/h^2}$
Answer	1
Solution	
Positive Marks	4
Negative Marks	1
Question	The dimension of coefficient of thermal conductivity is $\left(\frac{dQ}{dt}=-KA\frac{dT}{dx}\right)$, where dQ is heat transferred, dt is time taken, K is coefficient of thermal conductivity, A is cross-sectional area of conductor, dT is change in temperature and dx is length of conductor.
Type	multiple choice
Option	$[MLT^{-2}K^{-1}]$
Option	$[MLT^{-3}K^{-1}]$
Option	$[MLT^{-2}K^{-1}]$
Option	$[MLT^{-3}K]$
Answer	2
Solution	
Positive Marks	4
Negative Marks	1
Question	A person performs an experiment and taken 200 readings. He repeats the same experiment and now takes 800 readings. By doing so
Type	multiple choice
Option	the probable error remains same
Option	the probable error is four times
Option	the probable error is halved
Option	the probable error is reduced by a factor ¼
Answer	4

Solution	
Positive Marks	4
Negative Marks	1
Question	Assertion (A): Absolute error in physical quantity can be positive, negative or zero. Reason (R): Absolute error is the magnitude of difference in measured value and true value of physical quantity. Choose the correct option:
Type	multiple choice
Option	Both A and R are true and R is the correct explanation of A.
Option	Both A and R are true and R is not the correct explanation of A.
Option	A is true but R is false.
Option	A is false but R is true.
Answer	4
Solution	
Positive Marks	4
Negative Marks	1